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WING FOR AN AIRCRAFT

Abstract

An aircraft wing, including a main wing with upper and lower skin panels, a leading-edge high lift assembly including a high lift body, and a connection assembly connecting the high lift body to the main wing, the high lift body being movable relative to the main wing between stowed and deployed positions. The connection assembly includes a rotation element mounted to both the high lift body and the main wing rotatably about a rotational axis. The upper skin panel has a leading-edge portion. The high lift body includes a rigid portion and a flexible skin portion with the rigid portion mounted to the rotation element. The flexible skin portion connects to the upper skin panel leading-edge portion and to the high lift body rigid portion, while being configured to be deformed between stowed and deployed deformation states, when the high lift body moves between the stowed and deployed positions.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application claims the benefit of European Patent Application Number 24162370.1 filed on Mar. 8, 2024, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a wing for an aircraft. Further aspects of the invention relate to a leading-edge high lift assembly for such a wing as well as an aircraft comprising such a leading-edge high lift assembly and/or such a wing.

BACKGROUND OF THE INVENTION

[0003] The wing comprises a main wing and a leading-edge high lift assembly movable relative to the main wing to increase lift of the wing. The leading-edge high lift assembly comprises a high lift body and a connection assembly. The high lift body is preferably a droop high lift body referred to as droop nose, droop leading-edge, droop flap or slat, in particular sealed slat. The connection assembly is configured for connecting the high lift body to the main wing, in particular to the leading-edge of the main wing, in such a way that the high lift body is movable relative to the main wing between a stowed position and a deployed position. The stowed position relates to a position where the wing profile has a lower curvature and might also be referred to as straight position, normal position, cruise position or retracted position, while the deployed position relates to a position where the wing profile has a higher curvature and might also be referred to as drooped position, landing position, on-ground position or extended position.

[0004] The connection assembly comprises at least one rotation element, such as a rotation rod or a link, that is mounted to the high lift body and that is mounted to the main wing rotatably about a first axis of rotation. The rotation element is mounted to the high lift body directly or indirectly, e.g., to a rib of the high lift body. The first axis of rotation preferably extends in parallel to the span direction and/or in parallel to the extension of leading-edge along the wing, so that the high lift body is rotated about the first axis of rotation when moved between the stowed position and the deployed position.

[0005] The main wing comprises an upper skin panel for contact with an ambient flow on an upper side of the main wing, and a lower skin panel for contact with the ambient flow on a lower side of the main wing. The upper skin panel has a leading-edge portion in the area of a leading-edge of the main wing and facing the high lift body. The upper skin panel and the lower skin panel might be joined at the leading-edge of the main wing, or might have an open end at the leading-edge of the main wing, where they might be connected or supported against each other via a front spar.

[0006] The high lift body comprises a rigid portion and a flexible skin portion configured for contact with the ambient flow. The rigid portion is mounted to the rotation element. The flexible skin portion is connected, preferably fixedly and directly connected, in particular directly attached or integrally formed, to the leading-edge portion of the upper skin panel and is connected, preferably fixedly and directly connected, in particular directly attached or integrally formed, to the rigid portion of the high lift body, preferably in the area of a leading-edge of the high lift body. The flexible skin portion is configured to be deformed, preferably elastically deformed, between a stowed deformation state and a deployed deformation state, when the high lift body is moved between the stowed position and the deployed position, so that the flexible skin portion is in the stowed deformation state when the high lift body is in the stowed position and the flexible skin portion is in the deployed deformation state when the high lift body is in the deployed position.

[0007] Similar wings are known in the art. By increasing the curvature of the wing profile when the

high lift body is moved to the deployed position, lift of the related aircraft can be increased, in particular to allow approach and landing with lower speed and on shorter runways. High lift assemblies with a drooping, downward rotating high lift body that is sealed to the leading-edge portion of the upper skin panel, such as droop nose assemblies, relate to simple and effective high lift devices. Some known droop nose assemblies have a step or discontinuity at the upper skin at the transition between the high lift body and the main wing when the high lift body is moved to the deployed position, which might lead to a peak in the aerodynamic pressure profile in this area. Other known droop nose assemblies have a complicated design to avoid discontinuities at the upper skin surface.

SUMMARY OF THE INVENTION

[0008] Therefore, an object of the present invention is to provide a wing having a very simple and efficient leading-edge high lift assembly with an advantageous aerodynamic pressure profile.

[0009] This object is achieved in that the rigid portion is mounted to the rotation element rotatably about a second axis of rotation. The second axis of rotation preferably extends in a span direction and/or in parallel to the first axis of rotation. In such a way, by rotation of the rotation element about the first axis of rotation, the high lift body can rotate relative to the rotation element about the second axis of rotation and can be pivoted downwards to increase the curvature of the wing in a very simple and reliable manner, while at the same time the flexible skin portion of the high lift body is continuously bent without forming any discontinuities or steps, thereby leading to an aerodynamically advantageous wing profile.

[0010] According to a preferred embodiment, the rotation element comprises at least one link element or link plate. The link element might have an elongate form extending at least between the first and second axes of rotation, while the link plate might have a triangular or other plate-like form. Such a link or link plate relates to a very simple and efficient rotation element.

[0011] According to a further preferred embodiment, the rotation element is mounted to a rib of the main wing, preferably to a front end of a rib of the main wing, rotatably about the first axis of rotation, preferably so that the high lift body, when moved from the stowed position to the deployed position, is moved at least partly under the rib by the rotating rotation element.

Preferably, the first axis of rotation intersects the rib, i.e., extends within the cross-section of the main wing. The rib is a strong structural part and extends forward in the chord direction, which makes it advantageous to support the rotation element.

[0012] According to a further preferred embodiment, the rotation element is arranged such that the first axis of rotation extends above the second axis of rotation, i.e., on a higher level as the second axis of rotation, with respect to a normal position of the aircraft of the ground. This applies at least when the high lift body is in the deployed position, preferably also when the high lift body is in the stowed position, i.e., for the entire movement path of the rotation element. Preferably, the rotation element is arranged such that when the high lift body is moved from the stowed position to the deployed position, the rotation element rotates from a forward extending orientation to a downward extending orientation. In such a way, a very simple and efficient kinematic is obtained.

[0013] According to a further preferred embodiment, the connection assembly comprises a drive unit mounted to the main wing, preferably to the rib of the main wing, and directly or indirectly coupled to the rotation element for driving the rotation element for rotation about the first axis of rotation to move the high lift body between the stowed position and the deployed position. By being coupled to the rotation element, the drive unit can efficiently move the high lift body without being directly coupled to the high lift body, so that the drive unit can be formed smaller and lighter.

[0014] In particular, it is preferred that the drive unit comprises a drive strut mounted to the rotation element rotatably about a strut axis spaced from the first axis of rotation and preferably extending in the span direction and/or in parallel to the first axis of rotation. By such a drive strut the high lift body can be driven very efficiently.

[0015] It is further preferred that the strut axis is arranged between the first axis of rotation and the

second axis of rotation, or within a triangular arrangement together with the first axis of rotation and the second axis of rotation, in particular when the rotation element is formed as a link plate. In such a way, a minimum actuation movement is required to move the high lift body between the stowed position and the deployed position.

[0016] Alternatively, it is preferred that the strut axis is arranged in a position opposite from the second axis of rotation with respect to the first axis of rotation, so that the first axis of rotation is arranged between the strut axis and the second axis of rotation, i.e., the rotation element extends in both directions from the first axis of rotation. In such a way, a minimum actuation force is required to move the high lift body between the stowed position and the deployed position.

[0017] It is further preferred that the drive unit comprises a rotating actuator including a rotating arm rotatably driven about a drive axis and mounted to the drive strut rotatably about an arm axis. The rotating actuator is preferably mounted to the main wing, preferably to the rib of the main wing. Such a rotating actuator allows a very simple and efficient actuation.

[0018] Alternatively, it is preferred that the drive unit comprises a linear actuator, such as an electromagnetic linear motor or hydraulic cylinder, mounted to the main wing, preferably to the rib of the main wing, rotatably about an actuator axis, preferably at its one end, and configured for linearly moving the drive strut, preferably at its opposite end. Such a linear actuator also allows a very simple and efficient actuation.

[0019] According to a preferred embodiment, the wing further comprises a lower cover panel for covering a gap between the lower rear end of the high lift body and a front end of a lower skin panel. Preferably, the lower cover panel is mounted to the main wing, preferably to the rib of the main wing, and/or to the high lift body by a cover linkage comprising at least one cover link, so that the lower cover panel is preferably movable to cover the gap while the high lift body is moved between the stowed position and the deployed position. By such a lower cover panel closing the gap between the lower rear end of the high lift body and a front end of a lower skin panel, a possibly smooth lower surface of the wing can be obtained, thereby essentially reducing drag.

[0020] In particular, it is preferred that the cover linkage comprises one or more cover links, preferably two spaced cover links, rotatably mounting the lower cover panel to the main wing, preferably to the rib of the main wing. Preferably, the cover linkage comprises a further cover link rotatably mounting the lower cover panel to the rotation element. In such a way, the lower cover panel can be moved by the rotation element, so that the position of the lower cover panel can be adjusted to cover the gap at any time during movement of the high lift body.

[0021] Alternatively, it is preferred that the cover linkage comprises at least one cover link rotatably mounting the lower cover panel to the main wing, preferably to the rib of the main wing, while the lower cover panel is rotatably mounted to the high lift body, or rotatably mounting the lower cover panel to the high lift body while the lower cover panel is rotatably mounted to the main wing, preferably to the rib of the main wing. In such a way, an alternative design is provided, whereby the lower cover panel can be moved by the high lift body, so that the position of the lower cover panel can be adjusted to cover the gap at any time during movement of the high lift body.

[0022] A further aspect of the invention relates to a leading-edge high lift assembly for the wing according to any of the afore-described embodiments. The leading-edge high lift assembly comprises a high lift body and a connection assembly for connecting the high lift body to a main wing in such a way that the high lift body is movable relative to the main wing between a stowed position and a deployed position. The connection assembly comprises a rotation element that is mounted to the high lift body and that is configured to be mounted to the main wing rotatably about an axis of rotation. The high lift body comprises a rigid portion and a flexible skin portion. The rigid portion is mounted to the rotation element. The flexible skin portion is configured to be connected to a leading-edge portion of an upper skin panel of the main wing and is connected to the rigid portion of the high lift body. The flexible skin portion is configured to be deformed between a stowed deformation state and a deployed deformation state, when the high lift body is moved

between the stowed position and the deployed position. The rigid portion is mounted to the rotation element rotatably about a second axis of rotation. Features and effects described above in connection with the wing apply vis-a-vis also to the leading-edge high lift assembly.

[0023] A further aspect of the invention relates to an aircraft comprising the wing according to any of the afore-described embodiments and/or comprising the leading-edge high lift assembly according to any of the afore described embodiments. Features and effects described above in connection with the wing and in connection with the leading-edge high lift assembly apply vis-a-vis also in the case of the aircraft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Preferred embodiments of the present invention are explained hereinafter in more detail by means of a drawing. The drawing shows in

[0025] FIG. 1 shows a perspective view of an aircraft according to the invention,

[0026] FIGS. 2a and 2b show a schematic cross-sectional view across the span direction of a wing according to a first embodiment of the invention with a high lift body in the stowed position (FIG. 2a) and in the deployed position (FIG. 2b),

[0027] FIGS. 3a and 3b show a schematic cross-sectional view across the span direction of a wing according to a second embodiment of the invention with a high lift body in the stowed position (FIG. 3a) and in the deployed position (FIG. 3b),

[0028] FIGS. 4a and 4b show a schematic cross-sectional view across the span direction of a wing according to a third embodiment of the invention with a high lift body in the stowed position (FIG. 4a) and in the deployed position (FIG. 4b),

[0029] FIGS. 5a and 5b show a schematic cross-sectional view across the span direction of a wing according to a fourth embodiment of the invention with a high lift body in the stowed position (FIG. 5a) and in the deployed position (FIG. 5b),

[0030] FIGS. 6a and 6b show a schematic cross-sectional view across the span direction of a wing according to a fifth embodiment of the invention with a high lift body in the stowed position (FIG. 6a) and in the deployed position (FIG. 6b),

[0031] FIGS. 7a and 7b show a schematic cross-sectional view across the span direction of a wing according to a sixth embodiment of the invention with a high lift body in the stowed position (FIG. 7a) and in the deployed position (FIG. 7b),

[0032] FIGS. 8a and 8b show a schematic cross-sectional view across the span direction of a wing according to a seventh embodiment of the invention with a high lift body in the stowed position (FIG. 8a) and in the deployed position (FIG. 8b),

[0033] FIGS. 9a and 9b show a schematic cross-sectional view across the span direction of a wing according to an eighth embodiment of the invention with a high lift body in the stowed position (FIG. 9a) and in the deployed position (FIG. 9b), and

[0034] FIGS. 10a and 10b show a schematic cross-sectional view across the span direction of a wing according to a ninth embodiment of the invention with a high lift body in the stowed position (FIG. 10a) and in the deployed position (FIG. 10b).

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0035] In FIG. 1 an embodiment of an aircraft 1 according to the present invention is illustrated. The aircraft 1 comprises a fuselage 3, wings 5, a vertical tail plane 7 and a horizontal tail plane 9. FIGS. 2a to 10b show in more detail several embodiments of the wings 5.

[0036] FIGS. 2a and 2b show a first embodiment of the wing 5 according to the invention. The wing 5 comprises a main wing 11 and a leading-edge high lift assembly 13 movable relative to the main wing 11 to increase lift of the wing 5. The leading-edge high lift assembly 13 comprises a

high lift body **15** and a connection assembly **17**. The high lift body **15** is a droop high lift body also referred to as droop nose, droop leading-edge, droop flap or slat, in particular sealed slat. The connection assembly **17** is configured for connecting the high lift body **15** to the leading-edge of the main wing **11** in such a way that the high lift body **15** is movable relative to the main wing **11** between a stowed position and a deployed position. The stowed position relates to a position where the wing profile has a lower curvature, while the deployed position relates to a position where the wing profile has a higher curvature.

[0037] The connection assembly **17** comprises at least one rotation element **19** that is mounted to the high lift body **15** and that is mounted to the main wing **11** rotatably about a first axis of rotation **21**. The first axis of rotation **21** extends in parallel to a span direction **29** and in parallel to the extension of the leading-edge along the wing **5**, so that the high lift body **15** is rotated about the first axis of rotation **21** when moved between the stowed position and the deployed position. The first axis of rotation **21** of the rotation element **19** extends inside of an outer mold line of the wing **5**, specifically through a rib **23** of the main wing **11**.

[0038] The main wing **11** comprises an upper skin panel **31** for contact with an ambient flow **34** on an upper side of the main wing **11**, and a lower skin panel **33** for contact with the ambient flow **34** on a lower side of the main wing **11**. The upper skin panel **31** has a leading-edge portion **35** in the area of a leading-edge of the main wing **11** and facing the high lift body **15**. The upper skin panel **31** and the lower skin panel **33** are connected and supported against each other via a front spar **39**.

[0039] The high lift body **15** comprises a rigid portion **30** and a flexible skin portion **32** configured for contact with the ambient flow **34**. The rigid portion **30** is mounted to the rotation element **19** rotatably about a second axis of rotation **22**. The second axis of rotation **22** extends in a span direction **29** and in parallel to the first axis of rotation **21**. The flexible skin portion **32** is fixedly and directly attached to the leading-edge portion **35** of the upper skin panel **31** and is fixedly and directly attached to the rigid portion **30** of the high lift body **15** in the area of a leading edge **41** of the high lift body **15**. The flexible skin portion **32** is configured to be elastically deformed between a stowed deformation state and a deployed deformation state, when the high lift body **15** is moved between the stowed position and the deployed position, so that the flexible skin portion **32** is in the stowed deformation state when the high lift body **15** is in the stowed position and the flexible skin portion **32** is in the deployed deformation state when the high lift body **15** is in the deployed position.

[0040] The rotation element **19** is mounted to a front end of a rib **23** of the main wing **11** rotatably about the first axis of rotation **21**, so that the high lift body **15**, when moved from the stowed position to the deployed position, is moved partly under the rib **23** by the rotating rotation element **19**. The rotation element **19** is arranged such that the first axis of rotation **21** extends above the second axis of rotation **22**, i.e., on a higher level than the second axis of rotation **22**, with respect to a normal position of the aircraft **1** of the ground. This applies at least when the high lift body **15** is in the deployed position, but in the present embodiment also when the high lift body **15** is in the stowed position, i.e., for the entire movement path of the rotation element **19**. The rotation element **19** is arranged such that when the high lift body **15** is moved from the stowed position to the deployed position, the rotation element **19** rotates from a forward extending orientation to a downward extending orientation.

[0041] In the embodiments shown in FIGS. **2a**, **4a**, **6a-10a**, the rotation element **19** is in the form of a link element **43**, while in the embodiments shown in FIGS. **3a** and **5a** the rotation element **19** is in the form of a link plate **45**. The link element **43** has an elongate form extending at least between the first and second axes of rotation **21**, **22**, while the link plate **45** has a triangular form.

[0042] In the embodiments shown in FIGS. **3a** to **6a**, the connection assembly **17** comprises a drive unit **47** mounted to the rib **23** of the main wing **11** and coupled to the rotation element **19** for driving the rotation element **19** for rotation about the first axis of rotation **21** to move the high lift body **15** between the stowed position and the deployed position. The drive unit **47** comprises a

drive strut **49** mounted to the rotation element **19** rotatably about a strut axis **51** spaced from the first axis of rotation **21** and extending in the span direction **29** and in parallel to the first axis of rotation **21**.

[0043] In the embodiment shown in FIGS. **3** and **5**, where the rotation element **19** is formed as a link plate **45**, the strut axis **51** is arranged within a triangular arrangement together with the first axis of rotation **21** and the second axis of rotation **22**. In the embodiments shown in FIGS. **4a** and **6a**, where the rotation element **19** is formed as a link element **43**, the strut axis **51** is arranged in a position opposite from the second axis of rotation **22** with respect to the first axis of rotation **21**, so that the first axis of rotation **21** is arranged between the strut axis **51** and the second axis of rotation **22**, i.e., the rotation element **19** extends in both directions from the first axis of rotation **21**.

[0044] In the embodiments shown in FIGS. **3a** and **4a**, the drive unit **47** comprises a rotating actuator **53** including a rotating arm **55** rotatably driven about a drive axis **59** and mounted to the drive strut **49** rotatably about an arm axis **60**. The rotating actuator **53** is mounted to the rib **23** of the main wing **11**.

[0045] In the embodiments shown in FIGS. **5a** and **6a**, the drive unit **47** comprises a linear actuator **61**, such as an electromagnetic linear motor or hydraulic cylinder, mounted to the rib **23** of the main wing **11** rotatably about an actuator axis **63** at its one end **65**, and configured for linearly moving the drive strut **49** at its opposite end **67**.

[0046] In the embodiments shown in FIGS. **7a** to **10a**, the wing **11** further comprises a lower cover panel **69** for covering a gap **71** between the lower rear end of the high lift body **15** and a front end of a lower skin panel **33**. The lower cover panel **69** is mounted to the rib **23** of the main wing **11**, and/or to the high lift body **15** by a cover linkage **77** comprising at least one cover link **79**, so that the lower cover panel **69** is movable to cover the gap **71** while the high lift body **15** is moved between the stowed position and the deployed position.

[0047] In the embodiments shown in FIGS. **7a** and **8a**, the cover linkage **77** comprises two spaced cover links **79**, rotatably mounting the lower cover panel **69** to the rib **23** of the main wing **11**. In the embodiment shown in FIG. **8a**, the cover linkage **77** comprises a further cover link **79** rotatably mounting the lower cover panel **69** to the rotation element **19**.

[0048] In the embodiment shown in FIGS. **9a** and **9b**, the cover linkage **77** comprises one cover link **79** rotatably mounting the lower cover panel **69** to the rib **23** of the main wing **11**, while the lower cover panel **69** is rotatably mounted to the high lift body **15**. In the embodiment shown in FIG. **10a**, the cover linkage **77** comprises one cover link **79** rotatably mounting the lower cover panel **69** to the high lift body **15** while the lower cover panel **69** is rotatably mounted to the rib **23** of the main wing **11**.

[0049] As shown in FIG. **1**, the leading-edge high lift assembly **13** comprises a further connection assembly **17'** spaced from the connection assembly **17** in the span direction **29** and preferably formed as the connection assembly **17**. At least two spaced connection assemblies **17**, **17'** are provided to carry each high lift body **15**. Each connection assembly **17**, **17'** might also comprise more than one rotation element **19**.

[0050] By the invention as described above, a wing **5** with a very efficient leading-edge high lift assembly **13** can be formed. By rotation of the rotation element **19** about the first axis of rotation **21**, the high lift body **15** can rotate relative to the rotation element **19** about the second axis of rotation **22** and can be pivoted downwards to increase the curvature of the wing **5** in a very simple and reliable manner, while at the same time the flexible skin portion **32** of the high lift body **15** is continuously bent without forming any discontinuities or steps, thereby leading to an aerodynamically advantageous wing profile.

[0051] While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In

addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. A wing for an aircraft, comprising a main wing, and a leading-edge high lift assembly comprising a high lift body, and a connection assembly connecting the high lift body to the main wing in such a way that the high lift body is movable relative to the main wing between a stowed position and a deployed position, wherein the connection assembly comprises at least one rotation element that is mounted to the high lift body and that is mounted to the main wing rotatably about a first axis of rotation, wherein the main wing comprises an upper skin panel and a lower skin panel, wherein the upper skin panel has a leading-edge portion, wherein the high lift body comprises a rigid portion and a flexible skin portion, wherein the rigid portion is mounted to the rotation element, wherein the flexible skin portion is connected to the leading-edge portion of the upper skin panel of the main wing and is connected to the rigid portion of the high lift body, and wherein the flexible skin portion is configured to be deformed between a stowed deformation state and a deployed deformation state, when the high lift body is moved between the stowed position and the deployed position, and wherein the rigid portion is mounted to the rotation element rotatably about a second axis of rotation.
2. The wing according to claim 1, wherein the rotation element comprises at least one link element or link plate.
3. The wing according to claim 1, wherein the rotation element is mounted to a rib of the main wing rotatably about the first axis of rotation.
4. The wing according to claim 1, wherein the rotation element is arranged such that the first axis of rotation extends above the second axis of rotation.
5. The wing according to claim 1, wherein the connection assembly comprises a drive unit mounted to the main wing and coupled to the rotation element for driving the rotation element for rotation about the first axis of rotation.
6. The wing according to claim 5, wherein the drive unit comprises a drive strut mounted to the rotation element rotatably about a strut axis spaced from the first axis of rotation.
7. The wing according to claim 6, wherein the strut axis is arranged between the first axis of rotation and the second axis of rotation, or within a triangular arrangement with the first axis of rotation and the second axis of rotation.
8. The wing according to claim 6, wherein the strut axis is arranged in a position opposite from the second axis of rotation with respect to the first axis of rotation, so that the first axis of rotation is arranged between the strut axis and the second axis of rotation.
9. The wing according to claim 6, wherein the drive unit comprises a rotating actuator including a rotating arm rotatably driven about a drive axis and mounted to the drive strut rotatably about an arm axis.
10. The wing according to claim 6, wherein the drive unit comprises a linear actuator mounted to the main wing rotatably about an actuator axis and configured for linearly moving the drive strut.
11. The wing according to claim 1, further comprising a lower cover panel for covering a gap between a lower rear end of the high lift body and a front end of the lower skin panel, wherein the lower cover panel is mounted to the main wing, to the high lift body, or to both the main wing and the high lift body by a cover linkage comprising at least one cover link, so that the lower cover panel is movable to cover the gap while the high lift body is moved between the stowed position

and the deployed position.

12. The wing according to claim 11, wherein the cover linkage comprises one or more cover links rotatably mounting the lower cover panel to the main wing, and wherein the cover linkage comprises a further cover link rotatably mounting the lower cover panel to the rotation element.

13. The wing according to claim 11, wherein the cover linkage comprises at least one cover link rotatably mounting the lower cover panel to the main wing while the lower cover panel is rotatably mounted to the high lift body, or rotatably mounting the lower cover panel to the high lift body while the lower cover panel is rotatably mounted to the main wing.

14. A leading-edge high lift assembly for the wing according to claim 1, comprising a high lift body, and a connection assembly for connecting the high lift body to the main wing in such a way that the high lift body is movable relative to the main wing between a stowed position and a deployed position, wherein the connection assembly comprises at least one rotation element that is mounted to the high lift body and that is configured to be mounted to the main wing rotatably about a first axis of rotation, wherein the high lift body comprises a rigid portion and a flexible skin portion, wherein the rigid portion is mounted to the rotation element, wherein the flexible skin portion is configured to be connected to a leading-edge portion of an upper skin panel of the main wing and is connected to the rigid portion of the high lift body, wherein the flexible skin portion is configured to be deformed between a stowed deformation state and a deployed deformation state, when the high lift body is moved between the stowed position and the deployed position, and wherein the rigid portion is mounted to the rotation element rotatably about a second axis of rotation.

15. An aircraft comprising the wing according to claim 1.

16. An aircraft comprising the leading-edge high lift assembly according to claim 14.
